

Project Overview

Objective: Develop a simple, functional system where a manager can create field tasks and request an AI agent to suggest optimal task assignments to workers.

Technology Stack:

- **Frontend:** Vue.js (Composition API)
 - **Backend:** Django + Django REST Framework
 - **AI Module:** Python-based mock or Open-source LLM interaction
 - **Database:** SQLite (local)
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Scope for One-Day Implementation

Area	Feature
Frontend (Vue.js)	Login page, Dashboard to view/create tasks
Backend (Django REST)	API for authentication, task CRUD, AI assignment trigger
AI Integration	Mock service suggesting worker assignments
Database	Simple SQLite database

User Story

- **Manager logs in.**
 - **Manager creates tasks** (description + location).
 - **Manager clicks "Ask AI to Assign".**
 - **System suggests assignments** (mapping tasks to workers).
 - **Assignments are displayed on the dashboard.**
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Architecture Diagram

Frontend (Vue.js)

- Login Page
- Dashboard (Task CRUD + AI Assignment)

Backend (Django REST)

- Auth API
- Task API

- AI Suggestion API

Database (SQLite)

- Users
- Tasks

AI Service

- Simple Mock Assignment Engine
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Folder Structure

```
smart-task-assistant/  
├── frontend/      (Vue.js App)  
├── backend/       (Django Project)  
│   ├── sfop/     (Main settings)  
│   └── api/       (Task management and AI integration)
```



Backend (Django) - Key Components

Models:

```
class Task(models.Model):  
    description = models.CharField(max_length=255)  
    location = models.CharField(max_length=255)  
    assigned_to = models.CharField(max_length=255, blank=True, null=True)
```

Views:

- POST /tasks/ - Create a task
- GET /tasks/ - List tasks
- POST /ai/assign/ - Request AI suggestion for assignments

Mock AI Function:

```
def suggest_task_assignment(tasks):  
    workers = ['Alice', 'Bob', 'Charlie']  
    result = {}  
    for i, task in enumerate(tasks):  
        result[task.id] = workers[i % len(workers)]  
    return result
```



Frontend (Vue.js) - Key Components

Pages:

- **Login.vue:** Basic login form (username/password, hardcoded acceptance).
- **Dashboard.vue:**

- List of existing tasks.
- Form to create a new task.
- Button to "Ask AI" for assignments.
- Table to display task assignments.

API Calls (Axios):

- POST /tasks/ to create tasks.
 - GET /tasks/ to fetch tasks.
 - POST /ai/assign/ to trigger assignment suggestion.
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1-Day Implementation Timeline

Time Block Action

2 hours	Django backend setup (models, serializers, basic API)
2 hours	Vue.js frontend setup (pages, basic Axios integration)
1 hour	Mock AI service implementation
1 hour	Frontend-backend integration testing
Remaining	Styling and adjustments



Outcome

- Manager can log in.
 - Manager can create and view tasks.
 - Manager can request "AI assignment".
 - See task-to-worker assignment suggestions.
 - Everything runs locally.
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Possible Future Improvements

- Real authentication (JWT).
- Real open-source LLM integration (using Ollama or HuggingFace models).
- Worker login and mobile view.
- Live updates using Redis.
- Full cloud deployment with Kubernetes and Terraform.

